

Jennifer Weir

jenniferpweir.com | contact@jenniferpweir.com | github.com/jennweir

Platform engineer with experience building and operating OpenShift and Kubernetes clusters across cloud and bare-metal environments. Skilled in infrastructure automation, CI/CD pipelines, and distributed systems operations, with real-world experience deploying highly available clusters using Terraform, Tekton, and GitOps.

SKILLS

Kubernetes, OpenShift, Terraform, Infrastructure as Code (IaC), ArgoCD, Tekton, GitOps, Ansible, GCP, Bare Metal Infrastructure, Linux, YAML, Python, Bash, C++, Git/GitHub

EXPERIENCE

Ford Motor Company July 2024 – Present
Kubernetes Platform Engineer Dearborn, MI

- Automated provisioning of GCP infrastructure using Terraform modules, streamlining OpenShift cluster build workflows via an internally developed configuration discovery and scaffolding tool named scafctl
- Provisioned 20+ OpenShift clusters across GCP and bare-metal environments while supporting a 160+ cluster platform, resolving DNS, load balancing, networking, and hardware issues across environments
- Extended the teams' on-premise OpenShift platform type=none cluster build Tekton pipeline to support bare-metal deployments via Ansible, including DNS validation tasks for required OpenShift records
- Reduced platform technical debt by spearheading in-place bare metal server replacements of 30+ datacenter nodes across multiple production-level clusters with zero negative impact to active customer workloads
- Assisted application teams migrating hundreds of namespaces across 15+ on-premise PCF clusters to GCP-backed OpenShift clusters, supporting a platform-wide migration initiative over the course of a year

Instructional Aide @ University of Michigan Ann Arbor August 2023 – December 2023
SI 301 | Models of Social Information Processing Ann Arbor, MI

- Led office hours to help students with course material and graded homework/exams on a weekly basis

Ford Motor Company May 2023 – July 2023
Software Engineer Intern | Ford Pro Insure Dearborn, MI

- Pioneered the development of an open-source proof of concept for automated testing, leveraging Selenium and Cucumber to implement behavior-driven development unit testing of SaaS Origami insurance platform
- Authored a whitepaper showcasing proof of concept, encouraging business partners to engage in software testing with Selenium, leading to faster feedback loops and reduced time-to-market for the software product

Northrop Grumman June 2022 – August 2022
Systems Test Engineer Intern | NASA Contractor Palmdale, CA

- Refactored mission planning software systems by optimizing legacy VBA code to improve the UAV pilot experience, decreasing error and increasing satisfaction during autonomous vehicle route planning process
- Certified by Jet Propulsion Laboratory to solder electrical/electronic assemblies like circuit boards and wires

PROJECTS

HomeLab | github.com/jennweir/HomeLab August 2024 – Present

- Built and operate a highly available Kubernetes cluster on Raspberry Pis using Kube-VIP and Longhorn
- Built and operate a HA Bare Metal OKD cluster on refurbished PCs with Rook Ceph for distributed storage

EDUCATION

Penn State University | World Campus January 2026 – Present

- Master of **Computer Science**, Concentration in **Distributed and Cloud Computing** GPA: N/A
- Relevant Coursework: Large-Scale Software Systems, Data Structures and Algorithms

University of Michigan | Ann Arbor, MI May 2024

- Bachelor of Science in **Computer Science**, Minor in **Complex Systems** GPA: 3.67/4.0
- Relevant Coursework: Operating Systems, Web Systems, Software Engineering, Computer Organization, Data Structures and Algorithms, Foundations of Computer Science, Discrete Math, Computer Vision